Answer key

Practice Test 6 (State/National level) · 50 questions · 33 with checked app keys

Each answer lists short steps, the vault note (Question Types folder) and the app keys. Every key line was run in the app and its screen value matched the computed answer (check_keys.py). Accept numeric answers within about 2% with the right unit.

Multiple-choice answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14
C	A	B	D	A	B	C	A	B	A	C	B	A	C

15
D

Section I: Multiple choice

1. C. Attract each other

1. Ampère's 1820 experiments: parallel currents in the same direction attract;
2. opposite directions repel. (History fact only: no magnetic calculations in 2027.)

Note: A1 History/Ampere
App: no app: fact or concept

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2. A. Hermann von Helmholtz and Gustav Kirchhoff

1. Hertz worked in Berlin under Helmholtz (he became his assistant) and attended Kirchhoff's lectures.
2. PhD in Berlin, 1880.

Note: A1 History/Hertz
App: no app: fact or concept

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3. B. Locksmith

1. Johann Wolfgang Ohm was a self-taught locksmith.
2. Traps: Boole's father was a shoemaker; Coulomb was a military engineer.

Note: A1 History/Ohm
App: no app: fact or concept

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4. D. Kirchhoff and Boole

1. ampere (A), coulomb (C), volt (V), ohm (Ω), hertz (Hz) are SI units.
2. Kirchhoff has laws, Boole has an algebra: no units.

Note: A1 History/Units Named After People
App: no app: fact or concept

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5. A. Charge on her body drains away slowly, so no spark (ESD) jumps into the chips

1. Walking and moving build static charge on the body (thousands of volts).
2. A spark into a chip can destroy it. The strap keeps the body near ground; the resistor (about 1 MΩ) makes the drain slow and safe.

Note: A2 Static Electricity and Coulomb/Static Hazards
App: no app: fact or concept

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6. B. Negative

1. A negative rod pushes more electrons from the knob down to the leaves.
2. More negative charge on the leaves: they repel more and spread farther.
3. A positive rod would pull electrons up, and the leaves would close.

Note: A2 Static Electricity and Coulomb/Charging Methods and Electroscopes
App: no app: fact or concept

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7. C. Wet skin has a much lower resistance, so a much larger current flows

1. I = V/R. Dry skin can be about 100 kΩ; wet skin can be about 1 kΩ.
2. The current (what does harm) can be 100 times larger.

Note: A3 DC and AC/Shock and Safety
App: no app: fact or concept

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8. A. Alkaline AA cell

1. Primary cells: alkaline, zinc-carbon, lithium coin cells.
2. Secondary (rechargeable): NiMH, NiCd, Li-ion, lead-acid.

Note: A3 DC and AC/Batteries and Cells
App: no app: fact or concept

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9. B. 1 W · s

1. Power is energy per time: 1 W = 1 J/s, so 1 J = 1 W · s.
2. 1 V/A = 1 Ω. 1 A · s = 1 C.

Note: A4 Current Voltage Resistance Power Energy/Units Prefixes and Sig Figs
App: no app: fact or concept

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10. A. Becomes 1.5 times as large

1. Before: 3R. After the short: 2R.
2. I = V/R: the current becomes 3R/2R = 1.5 times as large.

Note: A5 DC Circuits/Open and Short Circuits
App: `N V1–0 9 1–2 10 s2–3 10 3–0 10` → `1–2 10: 900 mW –> 2.025 W x2.25 brighter` *(checked)*

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11. C. Its contacts conduct when it is NOT pressed, and open while it is pressed

1. Normally closed = closed (conducting) at rest.
2. The closed door presses the button: the contacts open and the light goes off.

Note: A6 Switches Relays Fuses Breakers GFCI/Switch Types
App: no app: fact or concept

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12. B. The highest voltage at which the fuse can safely break the circuit (stop the arc) when it blows

1. The current rating (5 A) sets when it blows.
2. The voltage rating is the most voltage it can interrupt safely. It can be used on any lower voltage.

Note: A6 Switches Relays Fuses Breakers GFCI/Fuses
App: no app: fact or concept

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13. A. The semiconductor material (its band gap)

1. Electrons fall across the band gap and give off a photon of that energy.
2. A bigger gap gives bluer light (and a larger forward voltage).

Note: A7 Diodes and LEDs/LEDs
App: no app: fact or concept

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14. C. 0.72 V

1. The diode current grows exponentially with voltage.
2. Doubling the current adds only about 18 mV (0.026 V × ln 2) at room temperature.
3. So about 0.72 V. It is not a resistor: the voltage does not double.

Note: A7 Diodes and LEDs/PN Junction and I-V Curve
App: no app: fact or concept

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15. D. Four 2-input AND gates

1. 7400 = quad NAND, 7402 = quad NOR, 7404 = hex NOT, 7408 = quad AND, 7432 = quad OR, 7486 = quad XOR.
2. Pin 7 is ground and pin 14 is V_{CC}.

Note: A8 Digital Logic/Logic Hardware and Chips
App: no app: fact or concept

Section II: Short answer and calculations

16. 5.1 N; x part +2.4 N, y part +4.5 N (up and to the right)

1. From q_2 (attraction, toward +x): 8.99e9 × 4.0e-6 × 6.0e-6 / 0.30^2 = 2.397 N.
2. From q_3 (repulsion, pushed away from q_3, toward +y): 8.99e9 × 4.0e-6 × 5.0e-6 / 0.20^2 = 4.495 N.
3. The two parts are perpendicular: use the Pythagorean theorem (no trig needed).
4. F = $\sqrt{(2.397^2 + 4.495^2)}$ = 5.094 N = 5.1 N.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: `Q 4u 0 0, −6u .3 0, 5u 0 −.2` → `F on q1: 5.093 N at 61.93 deg` *(checked)*

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17. 6.1 × 10^3 N/C, toward +x

1. From +6.0 nC (0.10 m away), pointing away from it (+x): 8.99e9 × 6.0e-9 / 0.10^2 = 5394 N/C.
2. From −3.0 nC (0.20 m away), pointing toward it (+x): 8.99e9 × 3.0e-9 / 0.20^2 = 674.2 N/C.
3. Both point toward +x, so they add: 6068 N/C = 6.1 × 10^3 N/C.

Note: A2 Static Electricity and Coulomb/Electric Field and Potential
App: `Q 6n 0, −3n .3 =.1` → `E at x=0.1: 6.067 kN/C right` *(checked)*

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18. 5.0 × 10^{13} electrons, from B to A (each sphere ends at −2.0 μC)

1. Total charge: +6.0 + (−10.0) = −4.0 μC. Shared equally: −2.0 μC each.
2. A goes from +6.0 to −2.0 μC: it gains 8.0 μC of negative charge, so electrons moved from B to A.
3. n = 8.0e-6 / 1.602e-19 = 5.0 × 10^{13}.

Note: A2 Static Electricity and Coulomb/Charge Basics
App: `a q8u` → `e Electrons = 4.9932E13 electrons` *(checked)*

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19. 10.0 V

1. Voltage lost inside: I r = 180 × 0.0150 = 2.70 V.
2. Terminal voltage = 12.7 – 2.70 = 10.0 V (subtraction: keep the tenths place).

Note: A5 DC Circuits/Internal Resistance and Power Transfer
App: `i e12.7 r.015 i180` → `v Terminal voltage = 10 V` *(checked)*

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20. 2.8 h

1. Usable charge: 50% of 7.0 Ah = 3.5 Ah.
2. Light current: I = P/V = 15 / 12.0 = 1.25 A.
3. Time = 3.5 Ah / 1.25 A = 2.8 h. (Same with energy: 3.5 Ah × 12.0 V = 42 Wh; 42 / 15 = 2.8 h.)

Note: A3 DC and AC/Batteries and Cells
App: `b c3.5Ah v12 p15` → `t Time (hours) = 2.8 h` *(checked)*

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21. (a) 16.7 ms; (b) 120 times

1. T = 1/f = 1/60 = 16.7 ms.
2. The sine wave crosses zero twice in each cycle (once going up, once going down).
3. 60 cycles × 2 = 120 times per second.

Note: A3 DC and AC/AC Characteristics
App: `w f60` → `t Period = 16.67 ms` *(checked)*

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22. Black = hot (120 V); white = neutral (return path, near 0 V); green or bare = ground (safety). The ground wire normally carries no current.

1. Hot (black) brings the current from the breaker panel.
2. Neutral (white) carries it back; it is joined to ground only at the panel.
3. Ground (green or bare copper) carries current only during a fault, to trip the breaker.

Note: A3 DC and AC/Household Wiring
App: no app: fact or concept

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23. 5.3×10^{-4} Ω (0.53 mΩ)

1. Area = 5.0 mm × 20.0 mm = 100 mm^2 = 1.0×10^{-4} m^2.
2. R = ρL/A = 2.65e-8 × 2.00 / 1.0e-4 = 5.30e-04 Ω = 530 μΩ (5.0 mm has 2 significant figures).

Note: A4 Current Voltage Resistance Power Energy/Resistance and Resistivity
App: `r p2.65*10^-8 l2 a5*20mm^2` → `r Resistance = 0.00053 ohm` *(checked)*

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24. 5.93 h; $6.40

1. Energy into the battery: 60% of 64.0 kWh = 38.4 kWh.
2. Energy from the wall: 38.4 / 0.900 = 42.67 kWh.
3. Time = 42.67 kWh / 7.20 kW = 5.93 h.
4. Cost = 42.67 × $0.150 = $6.40.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `s64*.6/.9/7.2` → `= 5.92593` *(checked)*
App: `e p7.2k h5.926 r.15` → `c Cost = $6.40 (6.4001)` *(checked)*

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25. (a) 75 kΩ ±10%: 67.5 kΩ to 82.5 kΩ; (b) 0.47 Ω and 1.0 Ω

1. (a) violet 7, green 5, orange ×1000: 75 000 ohm. Silver = ±10%: ±7.5 kohm.
2. (b) On an SMD code, R marks the decimal point: R47 = 0.47 ohm, 1R0 = 1.0 ohm.

Note: A5 DC Circuits/Resistor Color Codes
App: `cvgos` → `range 67.5k to 82.5k ohm` *(checked)*
App: `cR47` → `SMD code R47 = 470 mohm` *(checked)*
App: `c1R0` → `SMD code 1R0 = 1 ohm` *(checked)*

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26. 98.5 kΩ

1. At full scale the whole meter has 10.0 V across it and 100 µA through it.
2. Total R = 10.0 / 100e-6 = 100 kΩ.
3. Series resistor = 100 k − 1.50 k = 98.5 kΩ.

Note: A5 DC Circuits/Meters and Measurement
App: `o v10 i100u` → `r Resistance = 100 kohm` *(checked)*
App: `s100k-1.5k` → `= 98500` *(checked)*

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27. 5.00 Ω

1. Call the corners 1, 2, 3, 4 and measure from 1 to 2.
2. By symmetry, corners 3 and 4 are at the same voltage, so the 3-4 resistor carries no current: remove it.
3. Left: 1-2 direct (10.0), 1-3-2 (20.0) and 1-4-2 (20.0), all in parallel.
4. 1 / (1/10 + 1/20 + 1/20) = 5.00 ohm (R/2).

Note: A5 DC Circuits/Hard Resistor Networks
App: `N1-2-3-4-1 10 1-3 2-4 =1-2` → `R(1-2) = 5 ohm` *(checked)*

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28. 8.0 mA; yes, it trips (more than the about 5 mA trip level)

1. I = V/R = 120 / 15 000 = 8.0 mA.
2. This current goes to ground, not back on the neutral, so hot and neutral differ by 8.00 mA.
3. A GFCI trips at about 5 mA (4-6 mA): it trips.

Note: A6 Switches Relays Fuses Breakers GFCI/GFCI
App: `o v120 r15k` → `i Current = 8 mA` *(checked)*

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29. 2 three-way switches (one at each end) and 3 four-way switches (between them)

1. Every multi-place circuit has exactly two 3-way switches, one at each end of the chain.
2. Each extra place needs one 4-way switch: 5 − 2 = 3.

Note: A6 Switches Relays Fuses Breakers GFCI/3-Way and 4-Way Switching
App: no app: fact or concept

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30. 215 Ω

1. While the zener regulates, the load has 6.20 V, so the series resistor has 14.00 − 6.20 = 7.80 V.
2. Its current is 7.80 / 270 = 28.89 mA. This is shared by the zener and the load.
3. A smaller load takes more current. The limit is when the load takes all of it (zener current = 0).
4. $R_{L,min}$ = 6.20 / 0.02889 = 215 Ω. Any smaller load pulls the voltage below 6.20 V.

Note: A7 Diodes and LEDs/Special Diodes
App: `N V1-0 14, 1-2 270, Z0-2 6.2, 2-0 x, V2=6.2` → `Any x from 214.6 ohm up` *(checked)*

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31. F = C + A′B′D

1. Use the don't care 6 as a 1: then 2, 3, 6, 7, 10, 11, 14, 15 is every cell with C = 1 (an octet): C.
2. Minterm 1 (0001) pairs with 3 (0011): A'B'D.
3. F = C + A'B'D.

Note: A8 Digital Logic/Boolean Algebra and Simplification
App: `t abcd m1,2,3,7,10,11,14,15 d6` → `F = A'B'D + C` *(checked)*

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32. 101010001 = 337

1. 11010110 = 214, 01111011 = 123.
2. 214 + 123 = 337.
3. 337 = 256 + 64 + 16 + 1 = 101010001.

Note: A8 Digital Logic/Binary Numbers and Addition
App: `x11010110+01111011` → `sum bin 101010001` *(checked)*

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33. $I_0 = C'$, $I_1 = C$, $I_2 = 1$, $I_3 = 0$

1. AB = 00: rows 0 (F = 1) and 1 (F = 0): F = C'.
2. AB = 01: rows 2 (0) and 3 (1): F = C.
3. AB = 10: rows 4 and 5 both 1: $I_2 = 1$.
4. AB = 11: rows 6 and 7 both 0: $I_3 = 0$.

Note: A8 Digital Logic/Multiplexers and Decoders
App: `t abc m0,3,4,5` → `F = A'BC + AB' + B'C'` *(checked)*
App: `t abc m0,3,4,5` → `Mux 4:1, select A,B: D0.. = C' C 1 0` *(checked)*

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34. F = A′B′C + A′BC′ + AB′C′

1. Exactly one 1: rows 001, 010, 100 (minterms 1, 2, 4).
2. No two of these rows are next to each other on a K-map (each pair differs in 2 bits), so no term simplifies.
3. F = A'B'C + A'BC' + AB'C'.

Note: A8 Digital Logic/Logic Design Word Problems
App: `t abc m1,2,4` → `F = A'B'C + A'BC' + AB'C'` *(checked)*

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35. 2 full charges (2.55 charges' worth of energy)

1. Compare energy (Wh), not mAh: the voltages differ.
2. Bank: 10.0 Ah × 3.70 V = 37.0 Wh. Usable: 85% = 31.45 Wh.
3. Phone: 3.20 Ah × 3.85 V = 12.32 Wh.
4. 31.45 / 12.32 = 2.55: only 2 FULL charges.

Note: A9 Hands-On and Meters/Task - USB Charger
App: `b c10000 v3.7` → `w Energy = 37 Wh` *(checked)*
App: `s37*.85/12.32` → `= 2.55276` *(checked)*

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36. R2 is open (broken inside, or a bad breadboard contact at R2)

1. No current, so the good resistors R1 and R3 have no voltage drop (V = IR = 0).
2. The whole 9.0 V appears across the gap: the open part is where the meter reads the full battery voltage.
3. So R2 (or its contacts) is open.

Note: A9 Hands-On and Meters/Task - Troubleshoot
App: no app: fact or concept

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37. 680 || (100 + 47) = 120.9 Ω

1. 100 + 47 = 147 ohm in series.
2. 147 in parallel with 680: 147 × 680 / 827 = 120.87 ohm.
3. Error = +0.73%: inside 1%.

Note: A9 Hands-On and Meters/Task - Target Resistance
App: `h 47 100 330 680=120` → `120.9 (+0.726%) 680||(100+47)` *(checked)*
App: `s680/(100+47)` → `Req = 120.9 ohm` *(checked)*

Section III: Circuit problems

Problem A: two batteries in parallel branches (Kirchhoff's laws) (Questions 38–41)

Three branches join node A (top) and node B (bottom). Left: R1 = 2.00 Ω and E1 = 12.0 V. Middle: R2 = 3.00 Ω and E2 = 9.00 V. Right: R3 = 4.00 Ω. Both batteries have their + plate at the TOP. Let I_1 and I_2 be the currents flowing UP through E1 and E2.

38. 1.85 A

1. KCL at A: the current down through R3 is $I_1 + I_2$.
2. Loop E1-R1-R3: 12.0 = 2.00 I_1 + 4.00($I_1 + I_2$), so 6 I_1 + 4 I_2 = 12.
3. Loop E2-R2-R3: 9.00 = 3.00 I_2 + 4.00($I_1 + I_2$), so 4 I_1 + 7 I_2 = 9.
4. Determinant 6 × 7 − 4 × 4 = 26. I_1 = (12 × 7 − 4 × 9)/26 = 48/26 = 1.85 A.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a6 b4 c12 d4 e7 f9` → `x x (I1 or V1) = 1.8462` *(checked)*
App: `N V1-0 12 1-2 2 V3-0 9 3-2 3 2-0 4` → `V1-0 12: 1.846 A out of +, gives 22.15 W` *(checked)*

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39. I_2 = 231 mA (up, out of the + plate): E2 gives power

1. I_2 = (6 × 9 − 12 × 4)/26 = 6/26 = 231 mA.
2. It is positive: current leaves E2 at its + plate, so E2 gives power (it is not charged).

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a6 b4 c12 d4 e7 f9` → `y y (I2 or V2) = 0.23077` *(checked)*
App: `N V1-0 12 1-2 2 V3-0 9 3-2 3 2-0 4` → `V3-0 9: 230.8 mA out of +, gives 2.077 W` *(checked)*

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40. V_{AB} = 8.31 V; I_3 = 2.08 A (down)

1. $I_3 = I_1 + I_2$ = 54/26 = 2.0769 A.
2. V_{AB} = R3 I_3 = 4.00 × 2.0769 = 8.31 V.
3. Check with one node equation: (V − 12)/2 + (V − 9)/3 + V/4 = 0 gives V = 108/13 = 8.31 V.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1-0 12 1-2 2 V3-0 9 3-2 3 2-0 4` → `V1 = 12 V V2 = 8.308 V` *(checked)*
App: `N V1-0 12 1-2 2 V3-0 9 3-2 3 2-0 4` → `R2-0 4: 8.308 V 2.077 A 2>0 17.25 W` *(checked)*

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41. 22.2 W

1. P = E1 × I_1 = 12.0 × 1.8462 = 22.2 W.
2. Check: E1 + E2 give 22.15 + 2.08 = 24.23 W; the resistors use 24.23 W.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1-0 12 1-2 2 V3-0 9 3-2 3 2-0 4` → `V1-0 12: 1.846 A out of +, gives 22.15 W` *(checked)*

Problem B: an unbalanced bridge (Questions 42–45)

A 9.00 V ideal battery feeds the bridge below. P is between R1 = 150 Ω (top) and R2 = 100 Ω (bottom). Q is between R3 = 220 Ω (top) and R4 = 330 Ω (bottom). R5 = 120 Ω joins P and Q.

42. -1.80 V: Q is 1.80 V higher than P

1. $V_P = 9.00 \times 100/250 = 3.60$ V.
2. $V_Q = 9.00 \times 330/550 = 5.40$ V.
3. $V_P - V_Q = -1.80$ V.

Note: A5 DC Circuits/Dividers and Bulb Brightness
App: `N V1–0 9 1–2 150 2–0 100 1–3 220 3–0 330 V2–3` → `V2 – V3 = –1.8 V` *(checked)*

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43. 5.77 mA from Q to P

1. Node P: $(V_P - 9)/150 + V_P/100 + (V_P - V_Q)/120 = 0$.
2. Node Q: $(V_Q - 9)/220 + V_Q/330 + (V_Q - V_P)/120 = 0$.
3. Solve: $V_P = 3.9462$ V, $V_Q = 4.6385$ V.
4. I = (4.6385 – 3.9462)/120 = 5.77 mA, from Q (higher) to P.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 9 1–2 150 2–0 100 1–3 220 3–0 330 2–3 120` → `R2–3 120: 692.3 mV 5.769 mA 3>2 3.994 mW` *(checked)*

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44. 168 Ω

1. Current in R1: (9 – 3.9462)/150 = 33.69 mA.
2. Current in R3: (9 – 4.6385)/220 = 19.83 mA.
3. Battery current = 53.52 mA. $R_{eq} = 9.00 / 0.05352 = 168$ Ω.
4. It is not series-parallel: R5 carries current, so Kirchhoff (or the node equations) is needed.

Note: A5 DC Circuits/Hard Resistor Networks
App: `N1–2 150 2–0 100 1–3 220 3–0 330 2–3 120 =1–0` → `R(1–0) = 168.2 ohm` *(checked)*
App: `N V1–0 9 1–2 150 2–0 100 1–3 220 3–0 330 2–3 120` → `V1–0 9: 53.52 mA out of +, gives 481.7 mW` *(checked)*

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45. 147 Ω

1. No current in R5 means a balanced bridge: R1/R2 = R3/R4.
2. R4 = R3 × R2 / R1 = 220 × 100 / 150 = 147 Ω.

Note: A5 DC Circuits/Hard Resistor Networks
App: `W a150 b100 c220` → `x Rx (bottom right) = 146.7 ohm` *(checked)*

Problem C: a logic circuit (Questions 46–48)

G1 = NAND(A, B), G2 = OR(B, C), and F = AND(G1, G2).

46. F = (AB)′(B + C) = A′B + B′C

1. F = (AB)'(B + C). De Morgan: (AB)' = A' + B'.
2. F = (A' + B')(B + C) = A'B + A'C + B'B + B'C = A'B + A'C + B'C (B'B = 0).
3. A'C is the consensus of A'B and B'C, so it can be dropped: F = A'B + B'C.

Note: A8 Digital Logic/De Morgan and Bubble Pushing
App: `t–(ab)(b+c)` → `F = A'B + B'C` *(checked)*

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47. F = Σm(1, 2, 3, 5)

1. A'B: rows 010, 011 (2, 3).
2. B'C: rows 001, 101 (1, 5).
3. Together: 1, 2, 3, 5.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `t-(ab)(b+c)` → `m(1,2,3,5)` *(checked)*

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48. I_0 = C, I_1 = A′

1. Put B = 0 in F = A'B + B'C: F = C. So I_0 = C.
2. Put B = 1: F = A'. So I_1 = A'.
3. Check: F = B'C + BA' is the 2-to-1 mux form.

Note: A8 Digital Logic/Multiplexers and Decoders
App: `t-b(c)+b(-a)` → `F = A'B + B'C` *(checked)*

Problem D: a diode network (Questions 49–50)

+10.0 V feeds R1 = 1.00 kΩ to node X. From X to ground: branch 1 is a silicon diode D1 (0.70 V when on) in series with R2 = 1.00 kΩ; branch 2 is a red LED (2.00 V when on) in series with R3 = 470 Ω.

49. 3.62 V

1. KCL at X (current in = current out): (10.0 – V)/1000 = (V – 0.70)/1000 + (V – 2.00)/470.
2. Solve this one equation: V_X = 3.62 V.
3. Check: V_X is above 2.00 V, so both branches carry current forward: both are on.

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: `N V1-0 10, 1-2 1k, D2-3, 3-0 1k, D2-4 2, 4-0 470` → `V1 = 10 V V2 = 3.623 V` *(checked)*
App: `N V1-0 10, 1-2 1k, D2-3, 3-0 1k, D2-4 2, 4-0 470` → `D2-4 2: ON 3.454 mA 2>4 6.907 mW` *(checked)*

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50. 3.45 mA

1. I = (V_X – 2.00)/470 = (3.6232 – 2.00)/470 = 3.45 mA.
2. Check: D1 takes 2.923 mA; sum 6.377 mA = R1 current 6.377 mA.

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: `N V1-0 10, 1-2 1k, D2-3, 3-0 1k, D2-4 2, 4-0 470` → `D2-4 2: ON 3.454 mA 2>4 6.907 mW` *(checked)*
